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### Tamper evident assembly with RFID for syringes

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#### Abstract

The present disclosure describes a syringe assembly including a syringe including a Luer connection and a tip, a plunger, a tip cap, a tamper evident cap disposed over the tip cap, a film including a frangible connection, and an RFID tag. The film secures the tamper evident cap to the Luer connection, where the frangible connection can break under a force applied to the tamper evident cap such that the tamper evident cap disengages from the Luer connection when the frangible connection breaks. The RFID tag is disposed on the syringe and has an integrated circuit and an antenna configured to send/receive signals between an RFID reader.

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## Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation-in-part of U.S. patent application Ser. No. 16/408,947, filed May 10, 2019, which is claims the benefit of U.S. Provisional Patent App. No. 62/772,461, filed Nov. 28, 2018, the disclosures of which are hereby incorporated by reference herein.

### TECHNICAL FIELD

(1) The present disclosure generally relates to tamper detection devices, and, more particularly, to a tamper evident cap and film with RFID capabilities for detecting tampering of syringes filled with a material.

### BACKGROUND

(2) Syringe assemblies are used to hold, transport, and deliver materials. For example, syringes are often utilized in medical environments to administer one or more medicinal materials. Syringe assemblies may differ in size, and their specific dimensions are dictated by the desired application and the specific material to be administered. In some instances, syringes may be pre-filled with one or more materials that are then dispensed from the syringe and combined with other elements.

(3) Many industrial applications require mechanisms that prevent tampering with a particular product. This is especially the case with syringes used in the medical profession, where it is important for medical staff and patients to be aware of any tampering with the syringe or the material contained therein. Existing technology for detecting and preventing tampering is often cumbersome, difficult to use, increases risks of injury to the user, and increases the likelihood of

contaminating the patient or the medical environment. Further, the addition of a tampering device can often require changes to be made to the manufacturing process of a syringe or its constituent components, which increases associated production costs and complexity of manufacturing.

(4) Therefore, there is a need for tamper evident assembly configured to be used with syringes having preexisting designs that are filled with a material.

(5) Some industrial applications require identification or tracking of devices and materials. Within the medical field, it is important for staff to be aware of usage parameters of various medical devices or materials for proper storage, handling, and administration to patients. Existing technology for tracking medical devices and materials often relies on manual control and monitoring, which increase the likelihood of human error during the process. The existing technology is not automated, and tracking parameters cannot be easily controlled over the lifetime of the device or material. Furthermore, implementing such control processes after the device or material has been manufactured and prepared requires additional steps to be performed by the medical staff, as well as implementation of separate equipment and operating protocols.

(6) Accordingly, there is a need for an identification and monitoring system configured to be used with medical devices, such as syringes filled with a medical material.

## SUMMARY

(7) An embodiment of the present disclosure is syringe assembly comprising a syringe having a barrel body that extends from a proximal end to a distal end and defines a chamber extending along an axial direction therethrough, a Luer connection at the distal end, and a tip extending from the distal end along the axial direction and defining an outlet in fluid communication with the chamber, where the chamber contains a material. The syringe assembly also includes a plunger received within the chamber of the syringe to create a fluid seal within the barrel body, a tip cap defining a central passage configured to receive a portion of the tip such that the tip cap creates a fluid seal over the outlet, and a tamper evident assembly. The tamper evident assembly includes a tamper evident cap disposed over the tip cap, where the tamper evident cap has a main body that defines a proximal end defining an opening, a distal end opposite the proximal end along the axial direction, an outer surface, and an inner surface opposite the outer surface that defines a passage configured to receive the tip cap. The tamper evident assembly further includes a film including a frangible connection and configured to secure the tamper evident cap to the Luer connection, where the frangible connection is configured to break under a force applied to the tamper evident cap such that the tamper evident cap is configured to disengage from the Luer connection when the frangible connection breaks. The tamper evident assembly further includes a radio-frequency identification (RFID) tag disposed on the syringe and configured to receive and store data related to the syringe assembly. The RFID tag includes an integrated circuit, configured to store electronic information thereon, and an antenna operatively connected to the integrated circuit and configured to receive a signal from an RFID reader to cause transmission of the electronic data to or from the integrated circuit.

(8) Another embodiment of the present disclosure is a method of labeling a syringe assembly. The method comprises providing a label configured to be affixed to a syringe. The label includes a radio-frequency identification (RFID) tag, and the RFID tag has an integrated circuit and an antenna operatively connected to the integrated circuit. The method also includes transmitting a first set of data to the RFID tag and storing the first set of data in a memory of the integrated circuit of the RFID tag. The method also includes verifying the first set of data on the RFID tag. The method further includes transmitting a second set of data to the RFID tag and storing the second set of data in the integrated circuit, the second set of data being different from the first set of data.

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## Description

## BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The present application is further understood when read in conjunction with the appended drawings. For the purpose of illustrating the subject matter, there are shown in the drawings exemplary embodiments of the subject matter; however, the presently disclosed subject matter is not limited to the specific methods, devices, and systems disclosed. In the drawings:
- (2) FIG. 1 illustrates a perspective view of a syringe assembly in accordance with an embodiment of the present disclosure;
- (3) FIG. 2 illustrates an exploded view of a distal portion of the syringe assembly shown in FIG. 1;
- (4) FIG. 3 illustrates a cross-sectional view of a distal portion of the syringe assembly shown in FIG. 1, taken along line 3-3 shown in FIG. 1;
- (5) FIG. 4 illustrates a perspective view of the distal portion of the syringe assembly shown in FIG. 1;
- (6) FIG. 5 illustrates a process flow diagram of a method of filling a syringe with a material according to an embodiment of the present disclosure;
- (7) FIG. 6 illustrates a process flow diagram of a method of filling a syringe with a material according to another embodiment of the present disclosure;
- (8) FIG. 7 illustrates a process flow diagram of a method of applying a tamper evident cap and film to a prefilled syringe according to an embodiment of the present disclosure;
- (9) FIG. 8 illustrates a perspective view of a portion of a syringe assembly with an RFID tag in accordance with an embodiment of the present disclosure;
- (10) FIG. 9A illustrates an exploded perspective view of an RFID tag in accordance with an embodiment of the present disclosure;
- (11) FIG. 9B illustrates a front perspective view of an RFID tag (shown in phantom) covered by a film in accordance with an embodiment of the present disclosure;
- (12) FIG. 10A illustrates a front perspective view of an RFID tag in accordance with an embodiment of the present disclosure;
- (13) FIG. 10B illustrates a schematic of an RFID tag in accordance with an embodiment of the present disclosure;
- (14) FIG. 11 illustrates a schematic of an RFID network in accordance with an embodiment of this disclosure; and
- (15) FIG. 12 illustrates a flow chart of a method of labeling a syringe assembly in accordance with an embodiment of this disclosure.

## DETAILED DESCRIPTION OF ILLUSTRATIVE EMBODIMENTS

- (16) Described herein is a syringe assembly **10** that includes a tamper evident assembly **180** including a tamper evident cap **200** and a film **250**. Certain terminology is used to describe the syringe assembly **10** in the following description for convenience only and is not limiting. The words “right,” “left,” “lower,” “upper,” “lower,” “proximal,” and “distal” designate directions in the drawings to which reference is made. The words “inner” and “outer” refer to directions toward and away from, respectively, the geometric center of the description to describe the syringe assembly **10** and related parts thereof. The words “axially” and “radially” refer to directions along the orthogonal axial and radial directions A, R, respectively. The terminology includes the above-listed words, derivatives thereof and words of similar import.
- (17) Referring to FIGS. 1-4, the syringe assembly **10** includes a syringe **100** having a barrel body **103**. The barrel body **103** can extend from a proximal end **100a** to a distal end **100b** along the axial direction A. The barrel body **103** can be molded from glass in one embodiment, though other materials are contemplated. For example, the barrel body **103** can be molded from cyclic olefin copolymer (COC), cyclic olefin polymer (COP), or various other plastic materials. In one embodiment, the barrel body **103** can comprise a substantially transparent material, such that a user of the syringe assembly **10** can monitor the material levels within the barrel body **103**, though

barrel bodies **103** having various levels of opacity are contemplated. The barrel body **103** is depicted as comprising a substantially cylindrical shape, though the present disclosure is not intended to be limited as such. The barrel body **103** can have an outer surface **103a** that extends from the proximal end **100a** to the distal end **100b** along the axial direction A, an inner surface **103b** opposite the outer surface **103a** that also extends from the proximal end **100a** to the distal end **100b** along the axial direction A, and a distal surface **103c** that extends substantially along the radial direction R at the distal end **100b** of the barrel body **103**. The inner surface **103b** of the barrel body **103** defines a chamber **109** that extends along the axial direction A from an opening **106** at the proximal end **100a** to the tip **126** at the distal end **100b**. The chamber **109** is configured to receive and store a material, such as a liquid, for dispensing through the tip **126**. The syringe assembly **10** further includes a Luer connection **140** at the distal end **100b** of the barrel body **103**, where the Luer connection **140** will be discussed further below. The syringe **100** also includes a flange **112** extending radially outwards from the proximal end **100a** of the barrel body **103**, where the function of the flange **112** will be described further below. Though depicted as defining an oval shape with two flat, oppositely positioned sides, the flange **112** can define other shapes as desired.

(18) The chamber **109** can be sized and configured to receive a plunger **50**, such that the plunger **50** is capable of sliding along the axial direction A through the chamber **109**. The plunger **50** can have a substantially cylindrical body **53**, though the shape of the body **53** will generally conform to the shape of the chamber **109**. The body **53** can be comprised of a substantially flexible material such as rubber, though other embodiments are contemplated where the plunger **50** comprises other materials. The plunger **50** can further include a plurality of ridges **56** extending radially outwards from the body **53**. As shown, the plurality of ridges **56** extend substantially circumferentially around the body **53** and are aligned and spaced apart along the axial direction A. However, the ridges **56** can comprise different sizes, shapes, and arrangements in other embodiments. The ridges **56** can function to engage the inner surface **103b** of the barrel body **103** of the syringe so as to create a fluid seal between the plunger **50** and the syringe **100**. As the plunger **50** moves distally through the chamber **109**, the plunger **50** can function to push material out of the chamber **109** through the tip **126**. Alternatively, as the plunger **50** moves proximally through the chamber **109**, the plunger **50** can function to draw material into the chamber **109** through the tip **126**.

(19) The plunger **50** can define a bore that extends into the body **53** from its proximal end. The bore can be configured to engage a plunger rod **25** by any suitable means, e.g., screw fit, simple interference fit, snap-fit, or barbed engagement. The plunger rod **25** extends from a proximal end **25a** to a distal end **25b** opposite the proximal end **25a** along the axial direction A. The plunger rod **25** can comprise a rod body **28** at its center, where the rod body **28** comprises an elongated, axially-extending rod. Connected to the rod body **28**, the plunger rod **25** can include a plurality of walls **31** extending radially outwards from the rod body **28**. As depicted, each of the walls **31** comprises multiple sections that each extend from the rod body **28** by various distances. However, other embodiments of the wall **31** are contemplated. For example, each of the walls **31** can define a substantially rectangular body that extends radially outwards from the rod body **28** and axially along the length of the rod body **28**. The plunger rod **25** is shown as including four walls **31**, where the walls **31** are arranged about the rod body **28** circumferentially spaced apart 90 degrees, such that the arrangement of walls **31** forms a substantially plus-shaped orientation. However, the plunger rod **25** can include more or less walls **31** in other embodiments, and thus other arrangements of walls **31** can define other shapes. Additionally, it is contemplated that the walls **31** can define other shapes or extend to different extents along the axial length of the rod body **28** or radially outwards from the rod body **28**. The walls **31** can function to provide stability and strength to the plunger rod **25**, while minimizing the cross-sectional footprint of the plunger rod **25** so as to reduce material requirements for the plunger rod **25**, thus reducing overall weight of the syringe assembly **10**.

(20) Once the plunger **50** and the distal end **25b** of the plunger rod **25** are inserted into the chamber

**109** of the syringe **100**, and the proximal end **25a** of the plunger rod **25** is located outside the chamber **109**, the plunger rod **25** can be used to control dispensing of the material from within the chamber **109**. In operation, movement of the plunger rod **25**, and thus the plunger **50**, distally through the chamber **109** along the axial direction **A** forces material to flow out of the chamber **109** through the tip **126**. To do this, a user can, using one hand, pull the flange **112** of the syringe **100** and a flange **37** of the plunger rod **25** towards each other. Conversely, movement of the plunger rod **25**, and thus the plunger **50**, proximally through the chamber **109** along the axial direction **A** draws material into the chamber **109** through the tip **126**. To do this, a user can, using one or two hands, push the flange **112** of the syringe **100** and the flange **37** of the plunger rod **25** away from each other.

(21) Now referring to FIGS. 2-3, a Luer connection **140** of the syringe assembly **10** will be described in greater detail. In some embodiments, the Luer connection **140** is a separate component that is configured to be assembled onto a syringe **100**, such as over the tip **126** of the syringe **100**. The Luer connection **140** can comprise a ring **142** that is made of a material flexible enough to allow for radially expansion in the outward direction under pressure exerted on an inner wall **144** of the ring **142** when the central opening **148** of the Luer connection **140** is placed over the tip **126** of the syringe **100**. The Luer connection **140** can include a plurality of ribs **143** that can extend radially outwards from the outer surface of the Luer connection **140**. The ribs **143** can be arranged circumferentially around the Luer connection **140** so as to provide a texture that allows a label **110** and/or a film **250** to more easily engage the Luer connection **140**, as will be described further below. Though one embodiment of the ribs **143** is shown, the present disclosure is not intended to be limited to such. In other embodiments, the outer surface of the Luer connection **140** is substantially smooth, i.e., lacking any ribs. In some embodiments, the inner surface of the Luer connection **140** comprises internal threads **149** configured to be engaged with outer threads **160** of a tip cap **150**. In other embodiments, the inner surface of the Luer connection **140** is smooth.

(22) In other embodiments, the Luer connection **140** is formed as an integral portion of the syringe **100**, e.g., during a molding process. In such embodiments, the Luer connection **140** can extend from the distal end **100b** of a syringe barrel body **103** along the axial direction **A**. At the center of the Luer connection **140** is included the tip **126** that extends from the distal end **100b** of the syringe barrel body **103** along the axial direction **A**. The tip **126** has an outer surface **126a** and an inner surface **126b** opposite the outer surface **126a**, where the inner surface **126b** defines a passage **132** that extends through the tip **126** to an outlet **135** of the tip **126**. As the passage **132** and the outlet **135** are in fluid communication with the chamber **109** of the syringe **100**, the passage **132** and the outlet **135** thus define the pathway for material being dispensed from the chamber **109** of the syringe **100**.

(23) In some embodiments, as depicted in FIG. 3, the assembly further comprises a label **110**. In certain embodiments, the label **110** is continuous from a proximal end **110a** to a distal end **110b** opposite the proximal end **110a** over at least a portion of the outer surface **103a** of the syringe **100** and until at least a portion of the outer surface of the Luer connection **140**. In other embodiments, the label **110** is continuous over at least a portion of the outer surface **103a** of the syringe **100**, the entire length of the outer surface of the Luer connection **140**, and at least a portion of the outer surface of the tip cap **150**. In such embodiments, the label **110** can be provided with an annular frangible connection, which must be broken in order to remove the tip cap **150** from the Luer connection **140**. In some embodiments, as depicted in FIG. 3, a portion of the label **110** is imprisoned between a film **250** and the syringe **100** and/or between the film **250** and the Luer connection **140**.

(24) The label **110** can be attached to the syringe **100** and the Luer connection **140** (and, optionally, tip cap **150**) by any suitable means. In some embodiments, the label **110** comprises a thermoplastic material and is attached to the syringe **100** and/or Luer connection **140** by heat-activated shrink wrapping. In certain embodiments, the label **110** comprises an adhesive and is attached to the



syringe **100** and/or Luer connection **140** by applying pressure. In other embodiments, the label **110** is attached to the syringe **100** and/or Luer connection **140** by a combination of heat-activated shrink wrapping and an adhesive. In certain embodiments, the label **110** maintains the Luer connection **140** blocked in rotation and in translation with respect to the syringe **100** when the label **110** is attached. A preferred arrangement of a syringe **100**, Luer connection **140**, tip cap **150**, and label **110** is described in U.S. Pat. No. 9,731,082, the contents of which are hereby incorporated by reference in its entirety.

(25) The label **110** can be printed with information relating to the material contained within the chamber **109** of the syringe **100**, e.g., active ingredient name, brand name, active ingredient concentration, volume, manufacturer, lot number, etc. The label **110** can also be bar coded with any combination of this information. The label **110** can be comprised of any suitable material, such as a thermoplastic material, paper, or a combination thereof. Thermoplastic materials that are suitable for a label **110** include, but are not limited to, polyvinyl chloride (PVC), polyethylene terephthalate (PET), oriented polystyrene (OPS), oriented polypropylene (OPP), polylactic acid (PLA) and mixtures thereof.

(26) Continuing with FIGS. **1-4**, the tip **126** extends from the distal end **100b** of the barrel body **103** along the axial direction A. As depicted, the tip **126** can take the form of a substantially hollow tube, though other embodiments of the tip **126** are contemplated. The tip **126** can slightly taper inwards as it extends axially away from the distal surface **103c**, or can alternatively define a substantially constant diameter. The tip **126** can have an outer surface **126a** and an inner surface **126b** that defines a passage **132** that extends through the tip **126**. The passage **132** can extend from the chamber **109** of the syringe **100** to an outlet **135** of the tip **126**. As the passage **132** and the outlet **135** are in fluid communication with the chamber **109**, the passage **132** and the outlet **135** thus define a pathway for material being dispensed from the chamber **109** of the syringe **100**. The passage **132** can take the form of a cylinder, a tapered tube, a stepped design, or any other suitable shape as desired.

(27) After the syringe **100** has been filled, the outlet **135** needs to be sealed so as to prevent material from leaking out of the chamber **109**. To do this, a tip cap **150** can be attached to the Luer connection **140** so as to seal the outlet **135**. The tip cap **150** can extend from a proximal end **150a** to a distal end **150b** opposite the proximal end **150a** along the axial direction A. As depicted, the proximal end **150a** defines an opening **158**, whereas the distal end **150b** is closed. The tip cap **150** can define a central passage **154** extending along the axial direction A into the tip cap **150** from the opening **158**. The tip cap **150** can further define a ridge **170** extending radially outwards from the outer surface of the tip cap **150**, and substantially continuously around the entirety of the perimeter of the tip cap **150**. In operation, the ridge **170** of the tip cap **150** abuts the distal end of the Luer connection **140** when the tip cap **150** has sealed the outlet **135**. However, in other embodiments, the ridge **170** may only extend partially around the perimeter of the tip cap **150**, or may not be present at all. Further, a plurality of ribs **166** can extend radially outwards from the outer surface of the tip cap **150** distal to the ridge **170**. The ribs **166** can be arranged circumferentially around the tip cap **150** so as to provide a texture for grasping the tip cap **150** by a user of the syringe assembly **10**. Though one embodiment of the ribs **166** is shown, the present disclosure is not intended to be limited to such.

(28) In operation, the tip cap **150** is configured to be attached to the tip **126** of the syringe **100** so as to create a fluid seal over the outlet **135**. To accomplish this, after the syringe **100** has been filled during assembly, the tip cap **150** can be screwed onto the Luer connection **140** such that the outer threads **160** of the tip cap **150** engage the internal threads **149** of the Luer connection **140**.

Alternatively, the tip cap **150** can be pushed into the Luer connection **140** with a force along the axial direction A, such that the tip cap **150** can be secured to the Luer connection **140** through an interference fit. When this is done, the tip **126** can be received within the central passage **154** of the tip cap **150** and continue to be disposed further through the central passage **154** until the tip **126**

contacts the distal-most surface of the central passage **154**, which indicates to the user that the tip cap **150** is completely secured to the syringe **100**. At this point, the outlet **135** of the tip **126** can be disposed at the distal-most location in the central passage **154**, thus creating a fluid seal over the tip **126**. The ridge **170** can contact the upper surface of the outer wall of the Luer connection **140** so as to limit the extent to which the tip **126** can be disposed in the central passage **154**. Once disposed into the Luer connection **140**, the outlet **135** can be located at a distal-most part of the central passage **154** and the tip **126** can engage the inner surface of the tip cap **150**, thus creating a fluid seal over the outlet **135**.

(29) Now referring to FIGS. **1-4**, the tamper evident assembly **180** of the syringe assembly **10** will be discussed in detail. The tamper evident assembly **180** can include a tamper evident cap **200** and a film **250**. The tamper evident cap **200** can include a main body **204** that extends from a proximal end **200a** to a distal end **200b** opposite the proximal end **200a** along the axial direction A. The main body **204** can be configured as a substantially hollow cylinder, through other shapes are contemplated, as the shape of the tamper evident cap **200** can vary according to the shape of the syringe **100** and/or tip cap **150** of the particular syringe assembly **10** of which it is a part. The tamper evident cap **200** can be formed through injection molding, and can comprise a plastic such as medical grade polypropylene, polycarbonate, or polyethylene terephthalate. However, other methods of forming the tamper evident cap **200** and other materials for forming the tamper evident cap **200** are contemplated. The main body **204** defines an outer surface **204a** and an inner surface **204b** opposite the outer surface **204a**, where the inner surface **204b** defines a passage **208** configured to receive the Luer connection **140** and the tip cap **150**. The distal end **200b** of the main body **204** can be closed, whereas the proximal end **200a** can define an opening **212**, where the passage **208** extends from the opening **212** along the axial direction A and terminates within the main body **204** at a location proximal to the distal end **200b**. However, it is contemplated that in other embodiments the distal end **200b** of the main body **204** can be open.

(30) The proximal end **200a** of the main body **204** can define a ridge **220** that extends radially outwards from the outer surface **204a** of the main body **204**. The ridge **220** can extend circumferentially around an entirety of the main body **204**, and can define the proximal-most portion of the tamper evident cap **200**. However, in other embodiments the ridge **220** can be located elsewhere on the tamper evident cap **200** or extend to different extends about the tamper evident cap **200**. The tamper evident cap **200** can also include a plurality of ribs **216** that extend radially outwards from the outer surface **204a** of the main body **204** distal to the ridge **220**, where the plurality of ribs **216** are positioned circumferentially about the outer surface **204a**. The ribs **216** can define substantially hemispherical shapes, and can be spaced apart equidistantly about the circumference of the tamper evident cap **200**. However, though one particular number and arrangement of ribs **216** is shown, other numbers and arrangements are contemplated. In operation, the ribs **216** can be configured to engage the film **250** so as to secure the film **250** to the tamper evident cap **200**, as will be described further below. Although the tamper evident cap **200** is depicted to have a plurality of ribs **216** extending circumferentially around an entirety of the main body **204**, other means for engaging with the film **250** are contemplated, such as alternative types of protrusions, patterns, and/or textures. In some embodiments, the type, pattern, and/or texture of the protrusions can be indicative of the type of material contained within the chamber **109** of the syringe **100**. This allows the user of the syringe assembly **10** to easily determine what material is within the syringe **100** and helps avoid incorrect medicaments from being mistakenly applied to a patient.

(31) In the depicted embodiment, the inner surface **204b** is substantially smooth. However, in other embodiments, the inner surface **204b** is textured or comprises one or more protrusions that extend radially inwards from the inner surface **204b**. In one embodiment, the inner surface **204b** comprises a plurality of ribs positioned circumferentially about the inner surface **204b** that are configured to be positioned between complementary ribs **166** positioned circumferentially about the outer surface

of the tip cap **150**.

(32) In operation, after the tip cap **150** is engaged with the Luer connection **140**, the tamper evident cap **200** can be disposed over the tip cap **150**. In some embodiments, when the tamper evident cap **200** is disposed over the tip cap **150**, the tamper evident cap **200** is spaced in an entirety from the tip cap **150**, the Luer connection **140**, and the syringe **100**. In other embodiments, when the tamper evident cap **200** is disposed over the tip cap **150**, the tamper evident cap **200** is at least partially engaged with the tip cap **150**, the Luer connection **140** and/or the syringe **100**. For example, when disposed over the tip cap **150**, the distal-most portion of the inner surface **204b** of the tamper evident cap **200** can contact the distal-most surface of the distal end **150b** of the tip cap **150**. In other embodiments, the inner surface **204b** of the tamper evident cap **200** can be engaged with the outer surface of the tip cap **150** and/or the outer surface of the Luer connection **140** along the axial direction A. Preferably, any contact between the tamper evident cap **200** and the tip cap **150** is such that the fluid seal over the outlet **135** is not compromised by the attachment of the tamper evident cap **200** and/or film **250** to the syringe **100** or by the disengagement of the tamper evident cap **200** from the syringe **100** upon breakage of the frangible connection **254** of the film **250**.

(33) In the depicted embodiment, the tamper evident cap **200** may not include any features so as to secure the tamper evident cap **200** to the tip cap **150**, the Luer connection **140** and/or the syringe **100**. As a result, the tamper evident assembly **180** can include a film **250** disposed over portions of the syringe assembly **10** to secure the tamper evident cap **200** to the Luer connection **140** and/or the syringe **100**. The film **250** can define a body that extends from a proximal end **250a** to a distal end **250b** that is opposite the proximal end **250a** along the axial direction A. In the depicted embodiment, the proximal end **250a** of the film **250** can be disposed around a portion of the barrel body **103** of the syringe **100**, while the distal end **250b** of the film **250** can be disposed around a portion of the tamper evident cap **200**. A portion of the film **250** between the proximal end **250a** and the distal end **250b** can also engage the tip cap **150** and/or the outer surface of the Luer connection **140**.

(34) In some embodiments, the film **250** is continuous through at least a distal portion of the syringe **100**, the entire Luer connection **140**, and at least a proximal portion of the tamper evident cap **200**. In certain embodiments, the film **250** may extend on the syringe **100** for a length equal or more than 5 mm, e.g., 10 mm, 15 mm, 20 mm, 30 mm, or more, as measured from the distal end **100b** of the syringe **100**. Alternatively, or additionally, the film **250** may extend onto the syringe **100** for a length equal to or more than 10%, e.g., 20%, 40%, 60%, or more of the length of the syringe barrel body **103**. In certain embodiments, the film **250** may extend on the tamper evident cap **200** for a length equal to or more than 2 mm, e.g., 3 mm, 4 mm, 5 mm, 6 mm, or more, as measured from the proximal end **200a**. Alternatively, or additionally, the film **250** may extend onto the tamper evident cap **200** for a length equal to or more than 10%, e.g., 20%, 40%, 60%, or more of the length of the tamper evident cap **200**. In other embodiments, the film **250** is continuous from at least a proximal portion of the Luer connection **140** to at least a proximal portion of the tamper evident cap **200**, but does not cover any portion of the syringe **100**. In yet other embodiments, the film **250** is continuous over the entire Luer connection **140** and at least a proximal portion of the tamper evident cap **200**, but does not cover any portion of the syringe **100**.

(35) In operation, after the tip cap **150** engages the Luer connection **140** and the tamper evident cap **200** is disposed over the tip cap **150**, the film **250** can be fitted over the tamper evident cap **200**, the Luer connection **140** and/or the syringe **100** so as to secure the tamper evident cap **200** to the Luer connection **140** and/or the syringe **100**. In particular, the ribs **216** that extend radially outwards from the outer surface of the tamper evident cap **200** can be configured to secure the distal end **250b** of the film **250** to the tamper evident cap **200**. Similarly, the ribs **143** that extend radially from the outer surface of the Luer connection **140** can be configured to secure a portion of the film **250** to the Luer connection **140**. In one embodiment, the engagement between the film **250** and the tamper evident cap **200**, the tip cap **150**, the Luer connection **140**, and/or the syringe **100** is formed

by shrink-wrapping the film **250** over the tamper evident cap **200**, the Luer connection **140**, and/or the syringe **100**. In some embodiments, the tamper evident cap **200** is disposed over the entirety of the Luer connection **140** such that the film **250** is secured to the tamper evident cap **200** and a distal end of the syringe **100**, but the film does not directly contact the Luer connection **140**. In such embodiments, the tamper evident cap **200** is secured to the Luer connection **140** indirectly, i.e., without any direct contact between the film **250** and the Luer connection **140**. Further, the film **250** can be adhesive-bonded to the tamper evident cap **200**. Additionally, the film **250** can be adhesive-bonded to the Luer connection **140** and/or the syringe **100**. In addition to the above-described methods of attaching the film **250** to the other components of the syringe assembly **10**, various other methods of attaching the film **250** may be utilized as desired.

(36) To disengage the tamper evident cap **200** from the tip cap **150** and the syringe **100**, the film **250** can include a frangible connection **254** positioned axially between the proximal and distal ends **250a**, **250b** of the film **250**. The frangible connection **254** is configured to be the portion of the syringe assembly **10** that indicates to a user whether the syringe assembly **10** has been tampered with. When the film **250** secures the tamper evident cap **200** to the tip cap **150** and the syringe **100**, the frangible connection **254** is configured to break under a force applied to the tamper evident cap **200**. This force can be a clockwise or counter-clockwise rotational force, or any other force as desired. When the frangible connection **254** breaks, the tamper evident cap **200** is configured to disengage from the tip cap **150**. As a result, the distal end **250b** of the film **250** can remain attached to the tamper evident cap **200** and disengage from the remainder of the syringe assembly **10**. If a user of the syringe assembly **10** sees that the frangible connection **254** of the film **250** is broken, the user knows that the material within the chamber **109** of the syringe **100** may have been tampered with. However, if the frangible connection **254** is intact, the user can be assured of a greatly reduced risk that the material has been tampered with. Further, breaking the frangible connection **254** can prevent the tamper evident cap **200** from being reattached to the tip cap **150**.

(37) As depicted, the frangible connection **254** can comprise a plurality of frangible bridges **258** positioned circumferentially around the body of the film **250**. The frangible bridges **258** can be positioned around an entirety of the circumference of the film **250**, such that when the frangible connection **254** breaks, the distal end **250b** of the film **250** can be completely separated from the proximal end **250a**. Each of the frangible bridges **258** can comprise a thin, elongate portion of the film **250** that defines a substantially constant width as they extend along the axial direction A. However, it is contemplated that each of the frangible bridges **258** can taper in width along the axial direction A. Further, the frangible bridges **258** can be equidistantly spaced about the circumference of the film **250**, though various other spacing is contemplated. The frangible connection **254** can also define a plurality of gaps **262** that extend through the film **250** from its outer surface to its inner surface. Each of the gaps **262** can extend circumferentially between two adjacent frangible bridges **258**. In the depicted embodiment, each of the gaps **262** defines a substantially rectangular shape. Though each of the gaps **262** is shown as having a particular design, each of the gaps **262** can vary in design and spacing along with the design and spacing of each frangible bridge **258**. The inclusion of the frangible bridges **258** and gaps **262** allows the film **250** to be easily broken at the frangible connection **254**. Though the frangible connection **254** is depicted to comprise a plurality of frangible bridges **258** positioned around an entirety of the circumference of the film, other methods of forming a frangible connection **254** are contemplated, such as tear strips or pull tabs.

(38) The film **250** can be made of a thermoplastic material selected from the group consisting of polyvinyl chloride (PVC), ethylene vinyl acetate (EVA), polyethylene terephthalate (PET), oriented polystyrene (OPS), oriented polypropylene (OPP), polylactic acid (PLA) and mixtures thereof. In certain embodiments, the film **250** is made of PVC. In some embodiments, the interior surface of the film **250** or a portion thereof further comprises an adhesive material, such as a glue or a heat-activated adhesive. In certain embodiments, the interior surface of the distal end **250b** of the film

**250** to be disposed around a portion of the tamper evident cap **200** comprises an adhesive material.

(39) The film **250** can be blank, or the film **250** can be printed with information relating to the material contained within the chamber **109** of the syringe **100**, e.g., active ingredient name, brand name, active ingredient concentration, volume, manufacturer, lot number, etc. The film **250** can be fully transparent, partially transparent, or substantially opaque. The film **250** can also include a color-coded portion **266** that is indicative of the type of material contained within the chamber **109** of the syringe **100**. This allows the user of the syringe assembly **10** to easily determine what material is within the syringe **100** and helps avoid incorrect medicaments from being mistakenly applied to a patient. In one embodiment, the color-coded portion **266** comprises a substantially solid band that defines at least a portion of the distal end **250b** of the film **250** and extends circumferentially around the film **250**. However, alternative placements, shapes, and sizes of the color-coded portion **266** are contemplated. The color of the color-coded portion **266** can be selected from a plurality of colors that each correspond to a different material. In one embodiment, the relationship between the color of the color-coded portion **266** and the material contained within the syringe **100** can conform to the labeling standards set by ASTM D4774, such that the color-coded portion **266** can be universally recognized and understood within any variety of medical environments. The standards set by ASTM D4774 are shown in the below table. The examples provided for each drug class are exemplary only and not meant to be exhaustive. Drugs that do not fit into the classes shown in Table 1 can be labeled with black printing on a white background according to ASTM D4774 standards. Exceptions are noted by the “A” superscript.

(40) TABLE-US-00001 TABLE 1 ASTM D4774 Standards Drug Class Examples Pantone Color  
Induction Agents Etomidate, Ketamine, Methohexital, Propofol, Yellow Thiamylal, Thiopental  
Benzodiazepines Diazepam, Midazolam Orange 151 Benodiazepine Receptor Flumazenil Orange 151 and White  
Diagonal Antagonist Stripes Muscle Relaxants Succinylcholine.sup.A Fluorescent Red 805 (Depolarizer) Muscle Relaxants (Non Atracurium, Cisatracurium, Mivacurium, Fluorescent Red 805 Depolarizer) Pancuronium, Rocuronium, Vecuronium Relaxant Antagonist Endophonium, Neostigmine, Pyridostigmine Fluorescent Red 805 and White (Non-Depolarizer) Diagonal Stripes Narcotics Alfentanil, Fentanyl, Hydromorphone, Blue 297 Meperidine, Morphine, Sufentanil, Remifentanil Narcotic Antagonists Levallorphan, Naloxone Blue 297 and White Diagonal Stripes Vasopressors Ephedrine, Norepinephrine, Phenylephrine, Violet 256 Epinephrine.sup.A Hypotensive Agents Hydralazine, Nitroglycerine, Nitroprusside, Violet 256 and White Diagonal Phentolamine, Trimethaphan Stripes Local Anesthetics Bupivacaine, Chloroprocaine, Lidocaine, Gray 401 Mepivacaine, Procaine, Ropivacaine, Tetracaine Anticholinergic Agents Atropine, Glycopyrrolate, Scopolamine Green 367 Beta Blockers Esmolol, Labetolol, Metoprolol White Background with Copper 876U Bar Across Drug Name Major Tranquilizers and Droperidol, Inapsine, Haloperidol, Salmon 156 Anti-Emetics Levomepromazine, Metoclopramide, Ondasetron .sup.A Printed against the background color as reversed plate letters with a black bar running from edge to edge of the film

(41) Though the film **250** is described as including a color-coded portion **266**, it is also contemplated that in other embodiments all or a portion of the main body **204** of the tamper evident cap **200** defines a color-coded portion that is indicative of the type of material within the chamber **109** of the syringe **100** in combination with or in place of the color-coded portion **266**. This can be done through molding the color-coded portion of the tamper evident cap **200** out of a material having the color that corresponds to the type of the material. Like the color-coded portion **266**, the color-coded portion of the tamper evident cap **200** can comprise a color selected from a plurality of colors that each correspond to a different material. In one embodiment, the color-coded portion of the tamper evident cap **200** can conform to the labeling standards set by ASTM D4774.

(42) Another embodiment of the present disclosure is a pharmaceutical product comprising a syringe assembly **10** and a secondary packaging system therefor. In some embodiments, the secondary packaging is a pouch, blister, flow wrapper, or bag. The secondary packaging can be

comprised of an oxygen, light, and/or moisture barrier material, such as high-density polyethylene (HDPE), ethylene/vinyl alcohol copolymer (EVOH), polypropylene (PP), polyethylene terephthalate (PET), polyethylene naphthalate (PEN), and polyamide (PA), metalized film, aluminum foil, oxide coated films, and combinations thereof. In certain embodiments, the secondary packaging system also comprises an oxygen absorber. The oxygen absorber can be a sachet, pouch, canister, capsule, sticker, or strip that is placed inside of the secondary packaging. Alternatively, or additionally, the oxygen absorber can be incorporated into the material of the secondary packaging. In some embodiments, the oxygen absorber is selected from the group consisting of reduced iron compounds, catechol, ascorbic acid and analogs thereof, metal ligands, unsaturated hydrocarbons and polyamides.

(43) Yet another embodiment of the present disclosure is a pharmaceutical product comprising a plurality of syringe assemblies **10** and a container therefor. In some embodiments, the container is a box, carton, case, package, tray, or tin. Optionally, one or more of the syringe assemblies **10** can be enclosed within a secondary packaging system before being placed into the container. In certain embodiments, each syringe assembly **10** enclosed within the container is filled with the same active ingredient. In other embodiments, each syringe assembly **10** enclosed within the container is filled with a different active ingredient from the same drug class, a different active ingredient from a different drug class, or any combination thereof. For example, the pharmaceutical product can comprise a plurality of syringe assemblies **10** enclosed with a container, wherein the two or more of the syringe assemblies **10** are filled with a different active ingredient from a first drug class, and one or more syringe assemblies **10** are filled with an active ingredient from a second drug class.

(44) Now referring to FIG. 5, a method **300** of filling the syringe **100** will be described. Method **300** begins with step **302**, which comprises receiving the syringe **100**, where the syringe **100** comprises the barrel body **103** extending from a distal end **100b** having a Luer connection **140** and a tip **126** to an open proximal end **100a**. As stated above, the barrel body **103** defines a chamber **109** that extends along the axial direction A therethrough. Further, the tip **126** is configured to be received within the central passage **154** of the tip cap **150**, such that the inner surface of the tip cap **150** creates a fluid seal over the tip **126** and the outlet **135**. Then, step **306** comprises filling the chamber **109** with the material through the proximal end **100a** of the syringe **100**. Next, in step **310**, the plunger **50**, which is connected to the plunger rod **25**, is disposed within the chamber **109**.

(45) After step **310**, step **314** involves placing the tamper evident cap **200** over the tip cap **150**. In an alternative embodiment, step **314** can be performed after the syringe **100** and tip cap **150** are received and before the chamber **109** is filled with material. As stated above, though the distal-most portion of the inner surface of the tamper evident cap **200** may contact the distal-most portion of the outer surface of the tip cap **150** when the tamper evident cap **200** is placed over the tip cap **150**, the contact between the tamper evident cap **200** and the tip cap **150** may not be sufficient to secure the tamper evident cap **200** to the tip cap **150**. Accordingly, in step **318** the film **250** can be attached to the tamper evident cap **200** and the Luer connection **140** such that the film **250** secures the tamper evident cap **200** to the Luer connection **140**. In step **318**, the film **250** can also be attached to the syringe **100**. As stated previously, the tamper evident cap **200** can include ribs **216** that promote a stronger engagement between the tamper evident cap **200** and the film **250**, and the Luer connection **140** can include ribs **143** that promote a stronger engagement between the Luer connection **140** and the film **250**. As a result, the film **250** can secure the tamper evident cap **200** to the Luer connection **140** and the syringe **100** until the frangible connection **254** of the film **250** is broken under a force applied to the tamper evident cap **200**, at which time the tamper evident cap **200** can disengage from the tip cap **150**.

(46) In one embodiment, step **318** includes shrink-wrapping the film **250** over the tamper evident cap **200** and the Luer connection **140**. However, other methods of attaching the film **250** to the tamper evident cap **200** and the Luer connection **140** are contemplated. To create a more secure engagement between the film **250** and the Luer connection **140**, method **300** can include step **322**,

which involves bonding the film **250** to the Luer connection **140** and/or the syringe **100** via an adhesive. Additionally, to create a more secure engagement between the film **250** and the tamper evident cap **200**, method **300** can include step **326**, which involves bonding the film **250** to the tamper evident cap **200** via an adhesive. However, both of steps **322** and **326** are optional, as they can be each performed in isolation, in conjunction with each other, or not at all. As stated above, the film **250** can also include a color-coded portion **266** that is indicative of the type of material contained within the chamber **109** of the syringe **100**. The color of the color-coded portion **266** can be selected from a plurality of colors that each correspond to a different material. In one embodiment, the relationship between the color of the color-coded portion **266** and the material contained within the syringe **100** can conform to the labeling standards set by ASTM D4774, such that the color-coded portion **266** can be universally recognized and understood within any variety of medical environments.

(47) In another embodiment, as shown in FIG. 6, a method **400** of filling the syringe **100** with a material begins with step **402**, which comprises receiving the syringe **100**. Then, step **406** comprises filling the chamber **109** of the syringe **100** with the material through the outlet **135** of the tip **126**. After step **406**, in step **410** the tip cap **150** is placed over outlet **135** of the tip **126** to create a fluid seal over the outlet **135**. Next, in step **414**, the tamper evident cap **200** is placed over the tip cap **150**. To secure the tamper evident cap **200** to the tip cap **150**, in step **418** the film **250** is attached to the tamper evident cap **200** and the syringe **100** as described above in the context of method **300**.

(48) In yet another embodiment of the invention, as shown in FIG. 7, a method **500** of applying the tamper evident cap **200** to the syringe **100**, which is prefilled, is shown. Method **500** begins with step **502**, in which the syringe **100** is received prefilled with a material and the tip cap **150** is already placed over the outlet **135** of the tip **126**. Then, the tamper evident cap **200** is placed over the tip cap **150** in step **506**. To secure the tamper evident cap **200** to the tip cap **150**, in step **510** the film **250** is attached to the tamper evident cap **200** and the syringe **100** as described above in the context of method **300**.

(49) The material contained within the chamber **109** of the syringe **100** in the syringe assemblies **10** typically is a liquid, which can be aqueous, non-aqueous, or a combination of aqueous and non-aqueous liquids. In some embodiments, the liquid is a diluent intended for mixing with an active ingredient prior to administration to a subject. Exemplary diluents include, but are not limited to, water, 0.9% saline, 5% dextrose, Ringer's lactate solution, and other pharmaceutically acceptable diluents. In other embodiments, the liquid is a pharmaceutical formulation comprising an active ingredient and, optionally, one or more excipients. Thus, the invention provides a pharmaceutical product comprising a syringe assembly according to the present invention, wherein the liquid is a pharmaceutical formulation. Suitable excipients include, but are not limited to, a tonicity modifier, antioxidant, buffer, pH adjuster, preservative, solubilizer, stabilizer, or a combination of any of the foregoing. A diluent or pharmaceutical formulation can take on any suitable physical form including, but not limited to, solution, suspension, emulsion, or dispersion.

(50) The active ingredient of the pharmaceutical formulation can be a therapeutic agent, a diagnostic agent, a nutrient, or a combination thereof. Examples of therapeutic agents include, but are not limited to anti-infectives, anesthetics, analgesics, anticoagulants, chemotherapeutics, hormones, anti-hypertensives, anti-inflammatories, antiemetics, bronchodilators, adrenergics, immunoglobulins, antipsychotics, antidepressants, and combinations thereof. Examples of diagnostic agents include, but are not limited to x-ray, MRI and ultrasound contrast agents, cholecystokinetics, vasodilators, and combinations thereof. Examples of nutrients include, but are not limited to, salts, carbohydrates, minerals, vitamins, lipids, and combinations thereof.

(51) In some embodiments, the active ingredient is a compound useful for pain management, muscle relaxation, sedation, and/or anesthesia. In certain embodiments, the active ingredient is an opioid, a benzodiazepine, a beta blocker, or an  $\alpha$ .sub.2-adrenergic receptor agonist. In particular

embodiments, the active ingredient is morphine, hydromorphone, hydrocodone, oxycodone, oxymorphone, codeine, buprenorphine, naloxone, naltrexone, fentanyl, remifentanyl, sufentanyl, alfentanyl, meperidine, rocuronium, vecuronium, midazolam, lorazepam, diazepam, neostigmine, atropine, glycopyrrolate, dexmedetomidine, cisastracurium, ropivacaine, lidocaine, propofol, ketamine, succinylcholine, or a combination of the foregoing.

(52) In other embodiments, the active ingredient is moxifloxacin, linezolid, levofloxacin, levetiracetam, vancomycin, cefepime, aztreonam, cefoxitin, ceftriaxone, cefazolin, cefotaxime, ceftazidime, gentamicin, oxacillin, nafcillin, penicillin, cefuroxime, ticarcillin, clavulanic acid, piperacillin, tazobactam, azithromycin, meropenem, ertapenem, tigecycline, micafungin, metronidazole, fluconazole, itraconazole, posaconazole, heparin, enoxaparin, dalteparin, theophylline, acetaminophen (paracetamol), ibuprofen, acetylcysteine, decitabine, azacitidine, docetaxel, pemetrexed, palonosetron, aprepitant, fosaprepitant, famotidine, amiodarone, nitroglycerin, nicardipine, clevidipine, dobutamine, esmolol, labetalol, metoprolol, somatropin, liraglutide, abaloparatide, semaglutide, teriparatide, degarelix, sumatriptan, epinephrine, ephedrine, vasopressin, methotrexate, testosterone, hydroxyprogesterone, or a combination of the foregoing.

(53) In yet another embodiment, as shown in FIG. 8, the syringe **100** can include one or more components for tracking, monitoring, and/or identifying the syringe assembly **10** and/or its contents. In some aspects, the film **250** disposed on the syringe **100** may include a radio frequency identification (RFID) label or tag **900**. Signal communication to or from an RFID label is a form of wireless communication that uses radio waves to identify and track objects.

(54) The RFID tag **900** may be configured to be affixed to the film **250** or, alternatively, may be part of a unitary component of the film **250**. It will be appreciated that the RFID tag **900** can be utilized with any of the syringe assemblies disclosed throughout this application. The RFID tag **900** should be dimensioned and arranged relative to the syringe **100** such as to not interfere with the described structure or operation of the tamper evident cap **200** of any of the embodiments disclosed herein. In some aspects, the RFID tag **900** may be affixed to the syringe **100** separate from the film **250**. In some aspects, an embodiment with the RFID tag **900** may be devoid of the film **250**. In some aspects, the RFID tag **900** may be affixed to the tamper evident cap **200**. In some aspects, the RFID tag **900** may be affixed to both, the syringe **100** and the tamper evident cap **200**.

(55) Referring to FIG. 9A, the RFID tag **900** may include a backing **904** configured to be received on the syringe **100**. The backing **904** may be secured to the syringe **100** via an adhesive or another suitable fixing mechanism. The backing **904** may include an adhesive thereon or, alternatively, may be configured to receive an adhesive between the syringe **100** and the backing **904**. In some aspects, the film **250**, as described throughout this application, may comprise the backing **904**. The backing **904** may be a separate component of the film **250** or, alternatively, at least a portion of the film **250** may be used as the entire backing **904** of the RFID tag **900**.

(56) The RFID tag **900** further includes an integrated circuit **908** that is connected to an antenna **912**. The integrated circuit **908** is configured to store electronic data therein and to communicate with the antenna **912**. Referring to FIG. 10B, the integrated circuit **908** may include a controller **909** and a memory **910**. The memory **910** may include non-volatile memory and is configured for read/write access to receive, store, and allow access to electronic data transmitted to or from the RFID tag **900**. The controller **909** is configured to read the electronic data from the memory **910** and to write electronic data to the memory **910**. The controller **909** is configured to cause the antenna **912** to transmit the electronic data from the memory **910** to a reader **930** and to receive electronic data from the antenna **912** into the memory **910**, as will be described in detail below. The antenna **912** is configured to communicate with the integrated circuit **908** and to transmit electronic data to and/or from the integrated circuit **908**. The antenna **912** may be configured to supply power to the integrated circuit **908** sufficient to actuate the read/write process on the integrated circuit **908** to initiate transmission of the electronic data.

(57) The integrated circuit **908** and/or the antenna **912** may be disposed on the backing **904**, for



example, opposite the syringe **100**. In some aspects, the integrated circuit **908** and/or the antenna **912** may be adhered to the backing **904**. In some aspects, the integrated circuit **908** and the antenna **912** may be secured relative to the syringe **100** by an external label **916**. The external label **916** may be configured to adhere to the backing **904** (or the film **250**) such that the integrated circuit **908** and the antenna **912** are held in friction fit between the external label **916** and the backing **904** (or the film **250**). In some aspects, the external label **916** may be configured to contact the syringe **100** directly without contacting the backing **904** or the film **250**. The external label **916** may include printed text, barcodes, graphics, or other visual identifiers thereon. Exemplary visual identifiers that have been described throughout this application with respect to the film **250** may also apply to the external label **916** and/or to the backing **904**. In use, the RFID tag **900** may be hidden from view under the external label **916** or by the film **250** (see FIG. **9B**). In some aspects, such as shown in FIGS. **9A** and **10A**, the integrated circuit **908** and the antenna **912** may be held together on an inlay layer **906**. It will be understood that the inlay layer **906** serves to retain the integrated circuit **908** and the antenna **912** and may not be required for any other purpose in the RFID tag **900**. With reference to embodiments disclosed throughout this application, in some embodiments, at least a portion of the RFID tag **900** (e.g., the integrated circuit **908** and the antenna **912**) may be disposed between the film **250** and the syringe barrel body **103**.

(58) In some aspects, an intermediate layer **920** may be disposed between the external label **916** and the integrated circuit **908** and antenna **912**. The intermediate layer **920** may provide structural and/or electromagnetic protection to the integrated circuit **908** and the antenna **912**. The intermediate layer **920** may be disposed adjacent the inlay layer **906**.

(59) In some aspects, the backing **904** may be disposed such that the integrated circuit **908** and the antenna **912** are disposed between the backing **904** and the syringe **100**. In such embodiments, the RFID tag **900** may include the additional external label **916** contacting the backing **904** opposite the integrated circuit **908** and the antenna **912**, or, alternatively, such embodiments may be devoid of an external label **916** entirely.

(60) The RFID tag **900** can be used to track and monitor the syringe assembly to which it is affixed, for example, the syringe assembly **10** as disclosed throughout this application. The RFID tag **900** can include active or passive RFID technology. In an active RFID system, the RFID tag includes a transmitter and a power source configured to activate the RFID tag and transmitter to broadcast an electronic signal with electronic data. The power source may include a battery or photovoltaic cell. In a passive RFID system, an RFID reader is configured to send a radio signal to the RFID tag. The antenna on the RFID tag can receive the signal and use that signal to actuate transmittal of electronic data to/from the integrated circuit. The RFID tag **900** described herein can be designed and configured for either active or passive operation. It will be understood that if the RFID tag **900** is an active RFID tag, a transmitter and power source would need to be operationally connected thereto. In preferred embodiments, as shown in FIG. **11**, the RFID tag **900** may be configured to interact with various components of a network **940**. The RFID tag **900** can be a passive RFID tag that is configured to receive electronic power from an RFID reader **930**. The reader **930** is configured to receive data from the RFID tag **900**. Although referred to as a “reader,” it will be understood that such a device may be configured to transmit data to the RFID tag **900**. As such, the reader **930** may also be described as a writer **930**. In some aspects, the reading and writing functionalities can be performed by a single device **930**, which will be referred to as the reader **930**. The RFID reader **930** is configured to provide the connection between the RFID tag **900** and an external system configured to send and/or receive information to and/or from the RFID tag **900**. The external system may include a personal computing device, a network server, or a computing cloud. The RFID reader **930** is configured to communicate with the RFID tag **900** (or with a plurality of RFID tags **900**) that are within the read/write range of operation. The RFID reader **930** is configured to perform various tasks, such as simple continuous inventorying, searching for RFID tags **900** that meet predetermined criteria, writing (or encoding) to selected RFID tags **900**, or other

electronic communications.

(61) The RFID tag **900** can be designed to operate within known frequency ranges for RFID communication systems, including low frequency (LF), high frequency (HF), and ultra-high frequency (UHF). The LF band can cover frequencies from about 30 KHz to about 300 KHz. For RFID purposes, LF devices typically utilize either 125 KHz or 134.2 KHz frequencies. The LF band can be configured to operate within a range of up to approximately 10 cm for reading and/or writing from and/or to the RFID tag **900**. The HF band can include frequencies in the range of from about 3 MHz to about 30 MHz and may be configured to operate within a range of up to approximately 100 cm. For RFID purposes, HF devices typically operate at 13.56 MHz. The UHF frequency band can cover frequencies from about 300 MHz to about 3000 MHz and may be configured to operate within a range of up to approximately 1200 cm. For RFID purposes, UHF devices typically operate at a range of between about 860 MHz to about 960 MHz (for passive RFID) and at about 433 MHz (for active RFID). In some embodiments, the RFID tag **900** can be configured to operate at a frequency within the UHF spectrum although it will be appreciated that this disclosure is not limited to any particular frequency range.

(62) The RFID tag **900** can receive, store, transmit, and/or otherwise make available information that is electronically stored therein, for example in the integrated circuit **908**. With continued reference to FIG. **11**, the reader **930** can further communicate with a processor **944** that is configured to communicate with data in a database **942** to provide instructions to the reader **930** to receive data from the RFID tag **900** or send data to the RFID tag **900**, as well as to process the received data based on predetermined instructions. The RFID tag **900** can include various information, including at least a product identifier, product serial number, lot number, expiration date, or a combination of any of the above. The information can be sent to or received from the RFID tag **900** via the reader **930**. In some embodiments, the RFID tag **900** can include at least all four of the product identifier, serial number, lot number, and expiration date. The serial number may be unique and may pertain to the individual syringe assembly or component thereof or, alternatively, to a particular type of assembly or medicament therein. The product identifier can refer to a graphical or alpha-numeric string of characters that represents the particular products to which the RFID tag **900** is attributed, for example, a syringe assembly **10** or a medicament within the syringe assembly **10**. The product identifier may include a Global Trade Item Number (GTIN). The GTIN is an identifier for trade items that can be referenced within, or compared to, a database. In aspects where the RFID tag **900** is intended to correspond to a medicament, the GTIN may include a National Drug Code (NDC) that corresponds to a particular medicament or medicament component, such as a specific pharmaceutical compound. The particular NDC associated with a respective RFID tag **900** can be included in the NDC Directory maintained by the United States Food and Drug Administration (FDA). It will be appreciated that the GTIN may include a drug code that is different from the NDC, for example one that corresponds to a drug code directory outside of the United States. The RFID tag **900** may be designed to comply with one or more national or international standards for labeling, such as GSI and/or ISO. The RFID tag **900** may include the same information stored electronically within the integrated circuit **908** and appearing visually on the backing **904** or the external label **916**.

(63) In some aspects, the RFID tag **900** may receive some or all of the information detailed above at substantially the same time or, alternatively, may receive data over a plurality of write cycles, in which the data is electronically transmitted from the reader **930** to the RFID tag **900**. In some aspects, the RFID tag **900** may be pre-encoded with the product identifier information and with the serial number prior to being applied to a syringe assembly, such as any of the assemblies described throughout this application. The RFID tag **900** may receive additional data, such as the lot number and expiration date, at a later time. In some aspects, some or all of the electronic data provided on the RFID tag **900** may be modified by the reader **930**.

(64) An exemplary encoding process **950** is depicted in FIG. **12**. It will be appreciated that the

RFID tag **900** may be encoded in a different process and that some of the steps described in the process **950** may be omitted, repeated, or changed in order relative to other steps. Referring to step **954**, the RFID tag **900** is placed in electronic communication with the reader **930**, and the reader **930** electronically transmits a first set of data to the RFID tag **900**. The first set of data may include the product identifier and/or the serial number. The first set of data may be associated with a database **942** stored on the reader **930** or stored on an external computing device, server, network, or cloud and accessible by the reader **930**. The database **942** is configured to communicate with the reader **930** via a processor **944**. The processor **944** may be disposed on or in the reader **930** or on a separate computing device. The processor **944** and the database **942** may be disposed on the same device or on different devices. The processor **944** is configured to electronically communicate with the database **942** and with the reader **930**. The electronic communication may be wired or wireless. The reader **930** is configured to transmit the signal to the antenna **912** of the RFID tag **900**. The controller **909** in the integrated circuit **908** is configured to receive the signal from the antenna **912**, encode or decode the signal according to preprogrammed instructions to generate storable data associated with the signal, and move the data associated with the signal to the memory **910** in the integrated circuit **908**.

(65) In step **958**, the RFID tag **900** is then be applied to the syringe assembly **10** or another embodiment disclosed herein. The RFID tag **900** can be applied to the syringe **100** as described above. In some aspects, this step may be performed prior to step **954**, such that the RFID tag **900** is affixed to the syringe **100** prior to the first set of data being transmitted to the RFID tag **900**. Alternatively, this step may be performed later in the process **950**, for example after the step **962** described below.

(66) In step **962**, the reader **930** is configured to be placed in electronic connection with the RFID tag **900** such that data can be transmitted between the RFID tag **900** and the reader **930**. The reader **930** receives data that is stored on the RFID tag **900**, for example the first set of data that may include the product identifier and/or the serial number that was transmitted to the RFID tag **900** in step **954**. The controller **909** accesses the data stored on the memory **910**, performed any necessary encoding or decoding step according to preprogrammed instructions, and sends the data to the reader **930** through the antenna **912**. The reader **930** communicates the received information with a processor **944** configured to relate the received information with predetermined instructions, such as with a threshold, algorithm, or other computer programmed instruction. The processor **944** is configured to compare the received information per the predetermined instructions to determine if the information meets predetermined criteria. The comparison may be done with a predetermined set of data (for example a “control” set of data). In some aspects, the received set of data may include the first set of data that was transmitted to the RFID tag **900** in step **954**, and the control set of data may include the same information as the first set of data. The comparison step may include determining if the received set of data matches the control set of data. If the information meets the criteria (e.g. if the received set of data matches the control set of data), the RFID tag **900** (and the assembly to which the RFID tag **900** is affixed) is considered to be “verified,” and the process can continue to the next step. If the information received by the reader **930** in this step does not meet the predetermined criteria, the RFID tag **900** and the related assembly are not verified, and additional checking or verification steps may be performed. For example, if the RFID tag **900** is not verified, this could mean that the RFID tag **900** is damaged or improperly affixed to the syringe **100**, in which case the RFID tag **900** may be replaced. In some aspects, lack of verification could mean that the first set of data transmitted to the RFID tag **900** in step **954** was not successful, in which case, step **954** may be repeated. In some aspects, the RFID tag **900** may have been tampered with, in which case the RFID tag **900** and/or any other component of the syringe assembly may be fixed or replaced.

(67) In step **966**, the reader **930** is configured to be placed in electronic connection with the RFID tag **900** (or remain in electronic connection with the RFID tag **900**). The reader **930** is configured to

transmit a second set of data to the RFID tag **900**. The second set of data may include the lot number and/or the expiration date of the product or assembly. In some aspects, step **966** may be performed a plurality of times to transmit individual portions of data separately, for example, performed a first time to transmit the lot number to the RFID tag **900** and performed a second time to transmit the expiration date. It will be appreciated that additional information may be electronically transmitted to, and stored in, the integrated circuit **908** of the RFID tag **900**.

(68) In step **970**, the reader **930** and/or the connected processor **944** may update a database with information pertaining to the RFID tag **900** processed in steps **954** to **966**, for example to update inventory.

(69) In some aspects, an additional step of printing visual indicators on the RFID tag **900** may be conducted. Although visual information does not affect the electronic reading or writing to the RFID tag **900**, visual information may provide a level of redundancy to ensure that the necessary information appears on the label.

(70) By using radio waves to encode the RFID tags **900** and then read information from those RFID tags **900**, a user can track and monitor inventory without a visual line of sight. Because RFID readers **930** can receive electronic data from a plurality of RFID tags **900** simultaneously, less time can be spent on verifying individual tags (as would need to be done if utilizing a different tracking technique, such as visual inspection or barcode scanning). Information related to the RFID tags **900** can be stored in the database **942** and can be easily accessed. In some aspects, the database **942** may be configured to be updated automatically when information received by the reader **930** is different from the information stored in the database **942** with respect to the particular RFID tag **900** (i.e. with respect to the particular product identifier or serial number assigned to the RFID tag **900** or the syringe assembly). This allows for real-time updating of the database when a product with an RFID tag **900** is accessed, moved, or tampered with. In some aspects, a plurality of products may be disposed within a storage container, such as a cabinet. As the product is placed into the storage container, the reader **930** receives information related to the product, and the database **942** is updated to account for that product. If the product is removed from the storage container, the reader **930** no longer receives information related to that product and its affixed RFID tag **900**. The database **942** may be updated to account for removal of that product.

(71) By relying on RFID technology, inventory can be monitored and tracked more easily to ensure sufficient amount of desired products. Similarly, by monitoring expiration date, the inventory can be adjusted to remove expired products and ensure unexpired products are maintained at the desired quantities. Digitally monitoring and storing inventory information can also reduce information loss that often accompanies manual record keeping. Furthermore, as the process can be automated for all products, individual user approach differences are eliminated to increase precision in inventory management. Additionally, time spent on inventorying, locating expired products, and locating recalled lots is decreased. Using RFID technology further allows for immediate updating of the inventory when a product is removed and reduces chances of human error, such as when a user forgets to update inventory upon adding or removing a product. Additionally, monitoring how each tagged product or assembly is used, how often, how much, and in which circumstances can help users determine future inventory needs and/or efficacy of treatments. Information written to the RFID tag **900** can be validated through cGMP to ensure high quality and accuracy.

(72) While various inventive aspects, concepts and features of the inventions may be described and illustrated herein as embodied in combination in the exemplary embodiments, these various aspects, concepts and features may be used in many alternative embodiments, either individually or in various combinations and sub-combinations thereof. Unless expressly excluded herein all such combinations and sub-combinations are intended to be within the scope of the present inventions. Still further, while various alternative embodiments as to the various aspects, concepts, and features of the inventions—such as alternative materials, structures, configurations, methods, devices and

components, alternatives as to form, fit and function, and so on—may be described herein, such descriptions are not intended to be a complete or exhaustive list of available alternative embodiments, whether presently known or later developed. Those skilled in the art may readily adopt one or more of the inventive aspects, concepts or features into additional embodiments and uses within the scope of the present inventions even if such embodiments are not expressly disclosed herein. Additionally, even though some features, concepts or aspects of the inventions may be described herein as being a preferred arrangement or method, such description is not intended to suggest that such feature is required or necessary unless expressly so stated. Still further, exemplary or representative values and ranges may be included to assist in understanding the present disclosure; however, such values and ranges are not to be construed in a limiting sense and are intended to be critical values or ranges only if so expressly stated. Moreover, while various aspects, features, and concepts may be expressly identified herein as being inventive or forming part of an invention, such identification is not intended to be exclusive, but rather there may be inventive aspects, concepts, and features that are fully described herein without being expressly identified as such or as part of a specific invention, the scope of the inventions instead being set forth in the appended claims or the claims of related or continuing applications. Descriptions of exemplary methods or processes are not limited to inclusion of all steps as being required in all cases, nor is the order that the steps are presented to be construed as required or necessary unless expressly so stated.

(73) While the invention is described herein using a limited number of embodiments, these specific embodiments are not intended to limit the scope of the invention as otherwise described and claimed herein. The precise arrangement of various elements and order of the steps of articles and methods described herein are not to be considered limiting. For instance, although the steps of the methods are described with reference to sequential series of reference signs and progression of the blocks in the figures, the method can be implemented in a particular order as desired.

## Claims

1. A syringe assembly, comprising: a syringe having a barrel body that extends from a proximal end to a distal end and defines a chamber extending along an axial direction therethrough, a Luer connection at the distal end, and a tip extending from the distal end along the axial direction and defining an outlet in fluid communication with the chamber, wherein the chamber contains a material; a plunger received within the chamber of the syringe to create a fluid seal within the barrel body; a tip cap defining a central passage configured to receive a portion of the tip such that the tip cap creates a fluid seal over the outlet; a tamper evident assembly, comprising: a tamper evident cap disposed over the tip cap, wherein the tamper evident cap has a main body that defines a proximal end defining an opening, a closed distal end opposite the proximal end along the axial direction, an outer surface, and an inner surface opposite the outer surface that defines a passage extending from the opening of the proximal end and configured to receive the tip cap; and a film including a frangible connection and configured to secure the tamper evident cap to the Luer connection; and a radio-frequency identification (RFID) tag configured to receive and store data related to the syringe assembly, wherein the RFID tag includes an integrated circuit, configured to store electronic information thereon, and an antenna operatively connected to the integrated circuit and configured to receive a signal from an RFID reader to cause transmission of at least part of the electronic information from the integrated circuit, and wherein the frangible connection is configured to break under a force applied to the tamper evident cap such that the tamper evident cap is configured to disengage from the Luer connection when the frangible connection breaks while maintaining the tip cap over the outlet of the tip following disengagement of the tamper evident cap.

2. The syringe assembly of claim 1, wherein the film is shrink-wrapped over a portion of the

tamper evident cap and at least a portion of the Luer connection.

3. The syringe assembly of claim 1, wherein the film includes a color-coded portion that comprises a color selected from a plurality of colors that each correspond to a different material.

4. The syringe assembly of claim 3, wherein the color of the color-coded portion corresponds to the material in accordance with ASTM D4774.

5. The syringe assembly of claim 1, wherein the film is adhesive-bonded to the tamper evident cap.

6. The syringe assembly of claim 1, wherein the film is adhesive-bonded to the Luer connection.

7. The syringe assembly of claim 1, wherein the tamper evident cap includes a plurality of ribs extending radially outwards from the outer surface and configured to secure the film to the tamper evident cap.

8. The syringe assembly of claim 1, wherein the Luer connection includes a plurality of ribs extending outwards from an outer surface and configured to secure the film to the Luer connection.

9. The syringe assembly of claim 1, wherein a portion of the main body defines a color-coded portion that comprises a color selected from a plurality of colors that each correspond to a different material.

10. The syringe assembly of claim 1, wherein the tamper evident cap is spaced in an entirety from the syringe when the tamper evident cap engages the tip cap.

11. The syringe assembly of claim 1, wherein a portion of the tamper evident cap is in contact with the tip cap.

12. The syringe assembly of claim 11, wherein the fluid seal over the outlet is not compromised when the tamper evident cap is in contact with the tip cap.

13. The syringe assembly of claim 1, wherein the material includes an active ingredient that is (a) a therapeutic agent selected from a group consisting of anti-infectives, anesthetics, analgesics, anticoagulants, chemotherapeutics, hormones, antihypertensives, anti-inflammatories, antiemetics, bronchodilators, adrenergics, immunoglobulins, antipsychotics, and antidepressants or (b) a diagnostic agent selected from a group consisting of x-ray, MRI and ultrasound contrast agents, cholecystokinetics, and vasodilators.

14. The syringe assembly of claim 1, wherein the material includes an active ingredient selected from a group consisting of an opioid, benzodiazepine,  $\alpha$ 2-adrenergic receptor agonist, beta blocker, morphine, hydromorphone, hydrocodone, oxycodone, oxymorphone, codeine, buprenorphine, naloxone, naltrexone, fentanyl, remifentanyl, sufentanyl, alfentanyl, meperidine, rocuronium, vecuronium, midazolam, lorazepam, diazepam, neostigmine, atropine, glycopyrrolate, dexmedetomidine, cisastracurium, ropivacaine, lidocaine, propofol, ketamine, succinylcholine, moxifloxacin, linezolid, levofloxacin, levetiracetam, vancomycin, cefepime, aztreonam, cefoxitin, ceftriaxone, cefazolin, cefotaxime, ceftazidime, gentamicin, oxacillin, nafcillin, penicillin, cefuroxime, ticarcillin, clavulanic acid, piperacillin, tazobactam, azithromycin, meropenem, ertapenem, tigecycline, micafungin, metronidazole, fluconazole, itraconazole, posaconazole, heparin, enoxaparin, dalteparin, theophylline, acetaminophen (paracetamol), ibuprofen, acetylcysteine, decitabine, azacitidine, docetaxel, pemetrexed, palonosetron, aprepitant, fosaprepitant, famotidine, amiodarone, nitroglycerin, nicardipine, clevidipine, dobutamine, esmolol, labetalol, metoprolol, somatropin, liraglutide, abaloparatide, semaglutide, teriparatide, degarelix, sumatriptan, epinephrine, ephedrine, vasopressin, methotrexate, testosterone, and hydroxyprogesterone.

15. The syringe assembly of claim 1, wherein the film includes the RFID tag.

16. The syringe assembly of claim 1, wherein the RFID tag is a passive RFID tag and is configured to receive an electronic signal from the RFID reader to initiate transmission of the at least part of the electronic information.

17. The syringe assembly of claim 1, wherein the RFID tag is configured to operate at a frequency between 300 MHz and 3000 MHz.

18. The syringe assembly of claim 1, wherein the electronic information includes at least one of a

product identifier, a serial number, a lot number, or an expiration date.

19. The syringe assembly of claim 1, wherein the RFID tag is disposed on the syringe.

20. The syringe assembly of claim 1, wherein the proximal end of the tamper evident cap is in contact with the barrel body.

21. The syringe assembly of claim 1, wherein the tamper evident cap is spaced in an entirety from the tip cap.

22. A tamper evident assembly, comprising: a tamper evident cap having a main body that defines a proximal end defining an opening, a closed distal end opposite the proximal end along an axial direction, an outer surface, and an inner surface opposite the outer surface that defines a passage extending from the opening of the proximal end and configured to receive a tip cap and a Luer connection of a syringe; a film including a frangible connection and configured to secure the tamper evident cap to the Luer connection; and a radio-frequency identification (RFID) tag within the film and configured to receive and store data related to the syringe, wherein the RFID tag includes an integrated circuit, configured to store electronic information thereon, and an antenna operatively connected to the integrated circuit and configured to receive a signal from an RFID reader to cause transmission of at least part of the electronic information from the integrated circuit, and wherein the frangible connection is configured to break under a force applied to the tamper evident cap such that the tamper evident cap is configured to disengage from the Luer connection when the frangible connection breaks while maintaining the tip cap sealing the passage following disengagement of the tamper evident cap.

23. The tamper evident assembly of claim 22, wherein the frangible connection extends circumferentially around the tamper evident cap.

24. The tamper evident assembly of claim 22, wherein the film includes a color-coded portion that comprises a color selected from a plurality of colors that each correspond to a different material.

25. The tamper evident assembly of claim 22, wherein the electronic information includes at least one of a product identifier, a serial number, a lot number, or an expiration date.

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